

PA4 : IR-drop Prediction

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1. Assignment Requirement

In this PA, you are asked to implement and train an IR-drop predictor using XGBoost

Input: Raw features of each cell in the circuit

Output: Predicted IR-drop values of each cell

Learning objective: minimize the error between predicted value and real IR-drop value

2. The Selection of Training and Validation Sets

I randomly selected one-fifth of the 80 data points, which amounts to 16 data points, as the validation set. I then used five-fold cross-validation and chose the best-performing fold as the final allocation method.

Fold1 valadation: 23, 26, 27, 31, 36, 38, 41, 42, 43, 45, 46, 49, 51, 52, 57, 60

Fold2 valadation: 3, 5, 6, 10, 14, 20, 28, 30, 33, 34, 37, 48, 54, 56, 61, 64

Fold3 valadation: 2, 7, 8, 13, 15, 17, 21, 25, 29, 32, 35, 40, 47, 58, 62, 79

Fold4 valadation: 1, 4, 9, 11, 12, 16, 18, 19, 22, 24, 39, 44, 50, 55, 59, 75

Fold5 valadation: 53, 63, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 76, 77, 78, 80

The data points not selected are used as training set in each fold.

Evaluation for each fold

	Fold1	Fold2	Fold3	Fold4	Fold5
MAE(mV)	21.12	21.44	21.64	21.07	21.36
MaxE(mV)	136.53	128.53	126.32	142.42	132.13
CC	0.68	0.68	0.68	0.66	0.67
NRMSE(%)	3.59	3.62	3.65	3.65	3.63

I ultimately choose the results from fold2 for the allocation of my training and validation sets.

K-Fold Cross Validation



3. Feature Selections

I first selected 10 features that I believed were more likely to affect IR-drop from a physical perspective, and then I sequentially added other features to compare and evaluate the results.

The 10 features include **x, y, w, h, Reff, SPR, Cell type, Pinternal, Ipeak, and Ttransition**.

I ultimately chose to use these 10 features, adding Tcinput, TCoutput, and TCinternal as the final selected features.

Feature description

Feature of target cell	Description	Input dependent?
x, y	Physical location (coordinate)	N
w, h	Width, height (dimension)	N
R_{eff}	Effective resistance	N
SPR	Shortest path resistance	N
$Cell\ type$	Cell type	N
P_{leak}	Leakage power	N
C_{load}	Loading capacitor	N
Pi_{c1}, Pi_{c2}, Pi_r	Equivalent π model	N
TC_{input}	Toggle counts of input	Y
TC_{output}	Toggle counts of output	Y
$TC_{internal}$	Toggle counts of internal connection	Y
$T_{arrival}$	Minimum arrival time	Y
$P_{internal}$	Internal power	Y
P_{switch}	Switching power	Y
$T_{transition}$	Transition time	Y
I_{peak}	Peak current	Y

Add the remaining features to the training sequentially

	The 10 selected features	Add Pleak	Add Cload	Add pic1	Add pic2	Add pir	Add pic1,pic2	Add Tcinput, TCoutput, TCinternal	Add Tarrival	Add Pswitch
MAE(mV)	22.13	22.46	21.32	21.49	21.26	21.38	21.15	21.44	21.45	22.14
MaxE(mV)	129.37	132.02	132.33	131.72	132.03	133.63	135.36	128.53	128.8	121.1
CC	0.68	0.67	0.69	0.69	0.68	0.69	0.68	0.68	0.68	0.68
NRMSE(%)	3.71	3.76	3.6	3.62	3.6	3.6	3.58	3.62	3.62	3.72

4. Parameter Settings

Depth : 8

Learning Rate : 0.05,

Evaluation Metric: rmse,

Objective Function: reg:squarederror,

num_round = 500

5. My Best Evaluation Results

```
(PA4) [s111521051@cad ~/PA4]$ python Sample_evaluation.py
Test set is: [80 81 82 83 84 85 86 87 88 89]
Testing MEMC_81.csv
Testing MEMC_82.csv
Testing MEMC_83.csv
Testing MEMC_84.csv
Testing MEMC_85.csv
Testing MEMC_86.csv
Testing MEMC_87.csv
Testing MEMC_88.csv
Testing MEMC_89.csv
Testing MEMC_90.csv
total time: 16.964109659194946
prediction done
all_cells_golden shape: (223829, 10)
MAE(mV): 21.44 mV
MaxE(mV): 128.53 mV
CC: 0.68
NRMSE(%): 3.62 %
```

6. Hardness of PA4

Compared to previous assignments, for PA4, I only need to adjust the training parameters and methods. The challenge this time lies in how to train for a better result. I adjust and compare the evaluations through a table, recording the results of each attempt. This approach proves to be more efficient.

7. My Suggestion

Overall, PA4 is much simpler this time. There's no need to write complex code anymore. Thank you, teacher.

8. Reference

K-Fold Valadation (https://blog.csdn.net/weixin_39832727/article/details/110550677).